Research Project Application Form

Graduate School of Natural Sciences

Use of this form is mandatory for all large research projects, notably final thesis work ("afstuderen"). It must be handed in by the student at the student desk ("studentenbalie"), Buys Ballot Building room 184b, prior to starting the project. For Theoretical Physics research projects, a copy of the completed form should be handed in to the ITF secretariat. (Minnaert Building room 402)

Student	
Name of MSc programme	Theoretical Physics
First and last name of student	Anastasios Bouzikas
Email address	tasbouz@gmail.com
Student number	4102398
Project supervisor (first examiner)	
Name and title (must be a Utrecht University staff member)	Dr. Umut Gürsoy
Faculty, department and Research group (chair)	Faculty of Science, ITP
Email address	u.gursoy@uu.nl
Daily supervisor **Fill out this section only if the project supervisor is not the daily supervisor!**	
Name and title	dr. B.S.E (Bert) Vercnocke
Affiliation	University of Amsterdam
Address	POSTBUS 94485 1090 GL Amsterdam
Telephone number	0205255745
Email address	bvercno1@uva.nl
Second examiner	
Name and title (must be a Utrecht University staff member)	A second UU staff examiner is only mandatory for research projects of 30 ECTS or more. For smaller research projects or internships, the second examiner can be replaced by an internal or external expert. Prof. dr. S.J.G Vandoren

Faculty and Research group (chair)	Faculty of science, ITP
Email address	s.j.g.vandoren@uu.nl
Research project	
Title	Gravitational Solitons from Topology.
Location	Please give the UU faculty and research group (chair), or the name of the host institute. In the case of a host institute, please also fill out the 'work placement agreement'. (Utrecht University, Weekly visits in university of Amsterdam)
Starting date	28/02/2015
Ending date	30/11/2015
Number of ECTS	45
Short description of the project, including aims	In General Relativity (GR), the Kerr black hole is the unique vacuum solution for a given mass and angular momentum. In supergravity, the low energy limit of string theory that extends GR, this need not be the case. There is another mechanism that can support matter from gravitational collapse: non-trivial topology. Many "black hole microstate geometries" are known that are supported by cohomological fluxes on topological cycles in the geometry. This mechanism has been described in detail in terms of Komar integrals and a Smarr formula recently by Gibbons and Warner, in the context of five dimensional supergravity. In this project the student will investigate this mechanism thoroughly. The aims are to: * survey background literature on black hole microstate geometries and understand the role played by fluxes and topol * comprehend the use of Komar integrals in GR and its extensions, and how this can lead to general Smarr formulas befitting the slogan "no solitons without horizons or topology" * apply known Smarr formulas with topological contributions in examples * extend the role of topology in new settings such as a different dimensionality of spacetime and other asymptotic

Agreements between student and supervisors	
Number of hours available for supervision	1 hour per week
Planning/timing of the supervision (e.g. 'weekly meetings')	Weekly meetings
Agreed student work load (e.g. full time, 4 days/week, etc.)	4-5 days per week
Student absence (holidays, courses, etc.)	Courses, Summer Holidays
Supervisor absence (holidays, conferences, etc.)	Holidays, conferences, other work-related absence (seminars, work visits)
Intermediate evaluations of student performance (specify at least two dates)	02/06/2015 (Tuesday) 15/09/2015 (Tuesday)
Presentations to be held	Final Presentation (defence), Intermediate Presentation (learing purposes)
Lab/group meetings to be attended	UvA: String seminar, Journal Club VU: String seminar, Advanced Graduate seminar
Other activities agreed upon	—
Primary assessment criteria (please see appendix for examples. Final presentation and written thesis must be part of the assessment)	Level: understanding of the underlying matter; amount of work in relation to number of ECTS Problem handling: Methods and techniques and their understanding Skills: Analytical skills, distinguishing matters of major and minor importance, independence, self-reflectiveness, planning Originality: new work or insights beyond known results *

Copyright

By signing this document, the student declares to transfer the copyright of any and all products, including the tangible and intellectual products, of the research project to Utrecht University. The rights of the student by scientific standards to be a co-author of publications or to be otherwise acknowledged are still recognized.
The student is allowed and must upload his final assessed thesis to the university publication archive IGITUR. At a later stage, the thesis may be made public via IGITUR, or its access may be restricted temporarily or indefinitely.

* Report/Thesis: literature/background, problem statement, style of writing, quality of English, report structure, textual structure, coherence/consistency, conclusions/recommendations, neatness

Oral Presentation: Presentation/public speaking skills, structure, discussion skills, presentational aids

Signatures	
Student signature and date **(Must be signed when form is handed in at the student desk.)**	27. 2. 2015
Project supervisor signature and date **(Must be signed when form is handed in at the student desk.)**	26. 2. 2015
Master's programme coordinator name, signature, and date **(Must be signed when form is handed in at the student desk.)**	Cristiane de Morais Smith 27. 2. 2015 27. 2. 2015
Examination board member name, signature and date **(Approval and signature will be obtained by student desk.)**	17-3-2015 Dr RSW van Wal

Appendix: an incomplete list of possible assessment criteria

- Level – Theoretical level, practical/experimental level, understanding of the underlying matter, amount of work in relation to number of ECTS
- Problem handling – Methods, techniques, design instrumentation, implementation, experiments
- Skills – Analytical skills, model building, distinguishing matters of major and minor importance, evaluation, independence, collaboration, self-reflectiveness, planning
- Effort and attitude
- Originality
- Relevance, valorization
- Report/Thesis – literature survey, problem statement, style of writing, quality of Dutch/English, report structure, textual structure, coherence/consistency, conclusions/recommendations, neatness
- Oral presentation – presentation/public speaking skills, structure, discussion skills, presentational aids